CHRIST (Deemed to be University)
School of Engineering and Technology
BTech in Robotics and Mechatronics Engineering

Drone Technologies and its Transformative Applications

CIA 3 – Project Activities

Comprehensive Thermal Instability Evaluation of Drone-Integrated Battery Module Architectures

Submitted by

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Date of Submission: March 9, 2026

Aim

To evaluate the thermal stability of a drone battery module under heat stress conditions and analyze lithium-ion battery behavior during abnormal operating conditions.

Project Statement

A drone battery module must be assessed for safety under heat-stress conditions.

- Simulate temperature rise during abnormal operation.
- Identify triggers for thermal runaway.
- Propose mitigation strategies.
- Prepare a safety assessment report.

Project Solution

Lithium-ion batteries are widely used in drone systems because of their high energy density and lightweight characteristics. However, under abnormal operating conditions such as excessive current, internal short circuits, or mechanical damage, these batteries may experience thermal instability.

Thermal instability can lead to thermal runaway, a rapid increase in battery temperature caused by exothermic chemical reactions. This condition can result in battery failure, fire hazards, and system damage.

To analyze the thermal behavior of the battery system, a mathematical model was developed using the thermal energy balance equation.

$$mc_p \frac{dT}{dt} = Q_{gen} - Q_{loss} \tag{1}$$

where m is battery mass, c_p is specific heat capacity, T is temperature, Q_{gen} is heat generated, and Q_{loss} is heat dissipated.

Simulation Results

Normal Operation vs Thermal Runaway

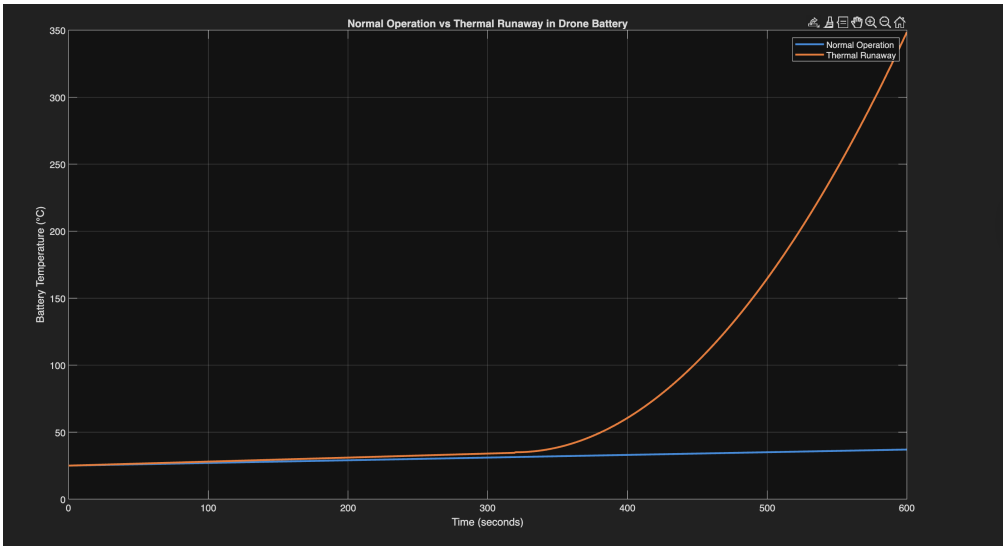

Figure 1: Comparison between normal battery operation and thermal runaway conditions.

Heat Generation vs Heat Dissipation

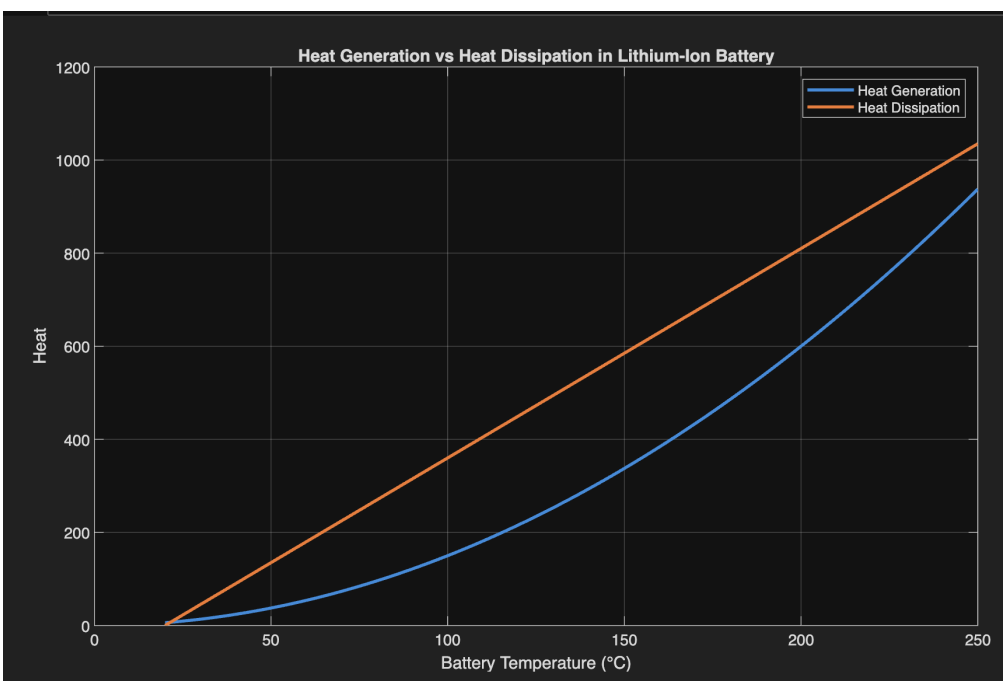

Figure 2: Heat generation compared with heat dissipation.

Battery Safety Limits

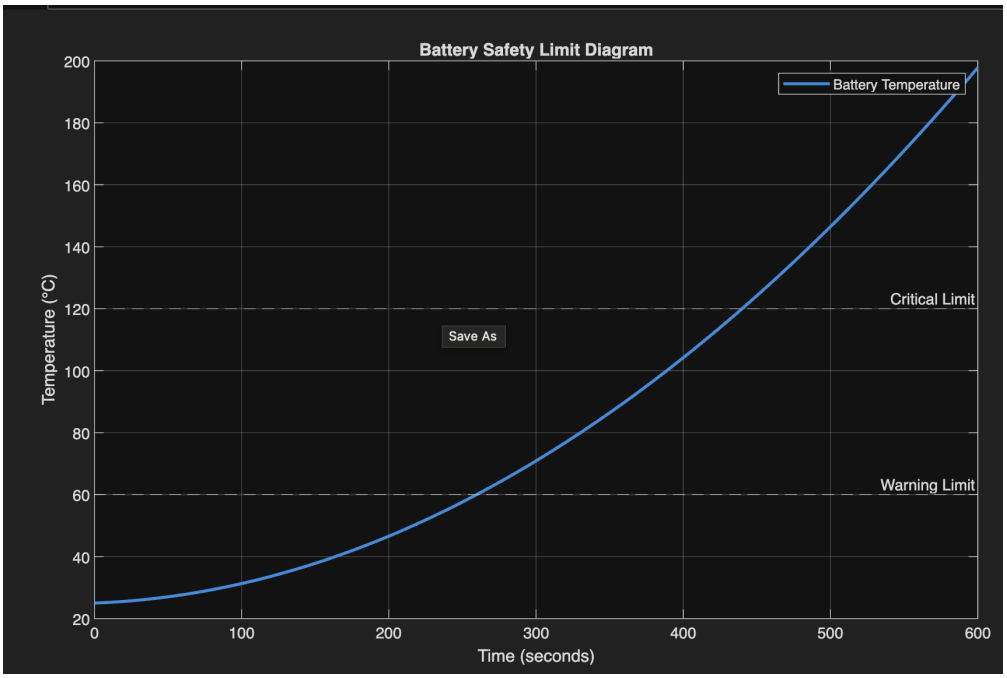

Figure 3: Battery temperature safety limits and thermal runaway thresholds.

Thermal Runaway Propagation

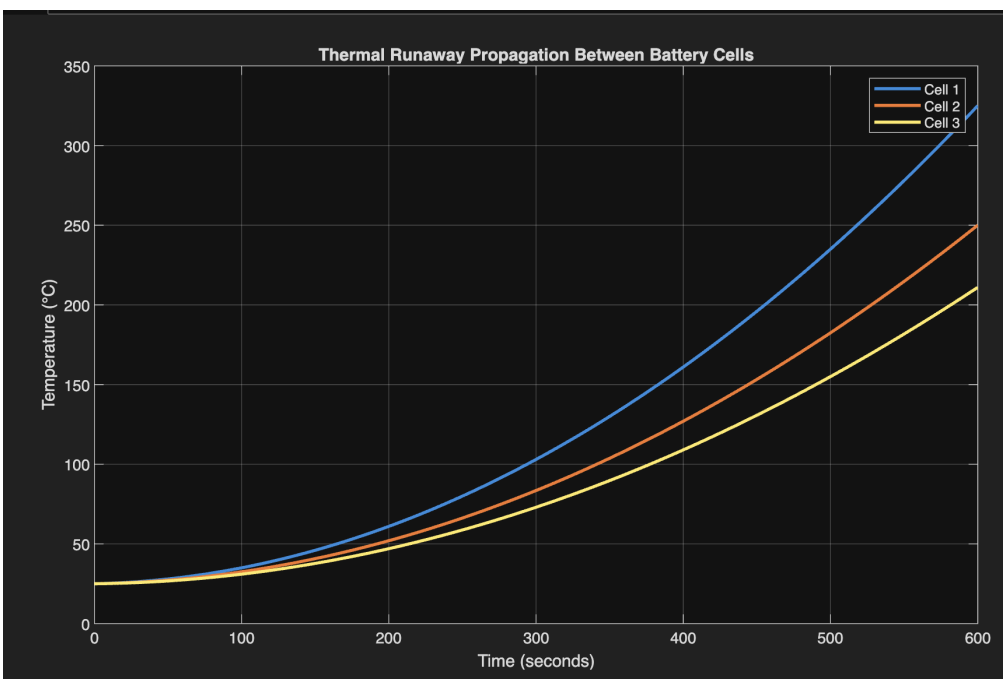

Figure 4: Thermal runaway propagation between battery cells.

Battery Voltage Profile

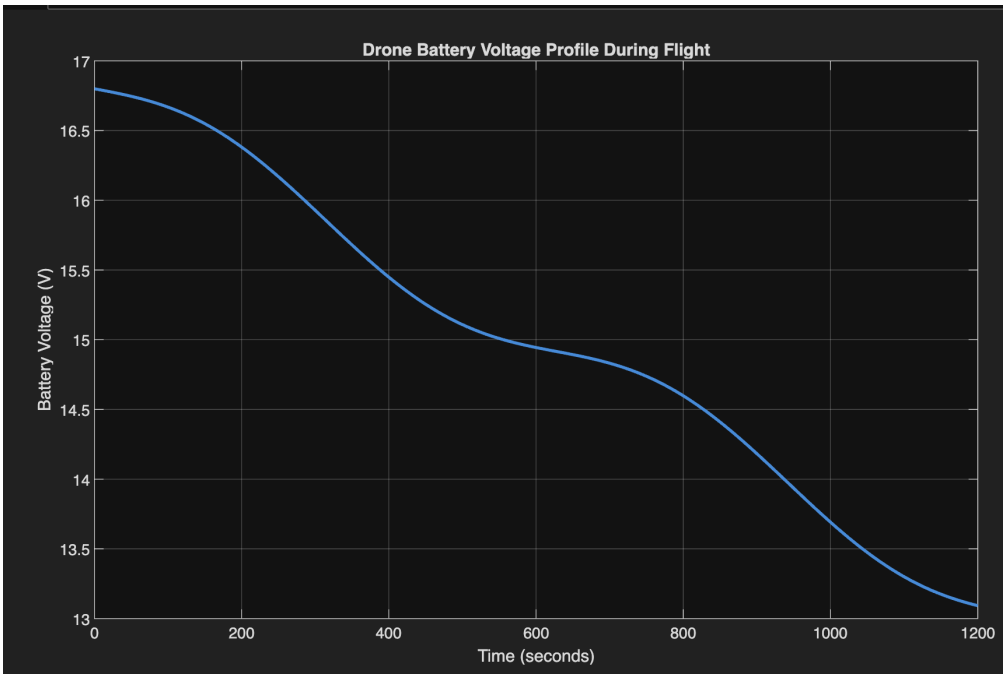

Figure 5: Voltage variation of drone battery during operation.

3D Temperature Distribution

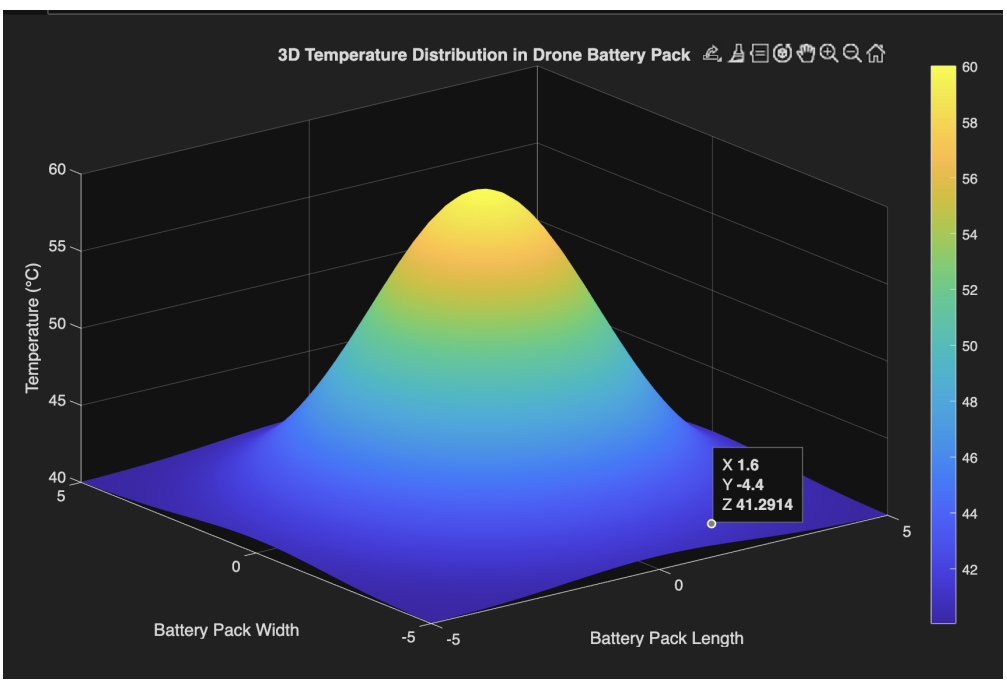

Figure 6: 3D temperature distribution across the drone battery pack.

Conclusion

This project analyzed the thermal instability behavior of lithium-ion batteries used in drone systems. The simulation results demonstrated that excessive current loads and inadequate heat dissipation can lead to rapid temperature rise and thermal runaway.

The study highlights the importance of thermal management systems, battery monitoring mechanisms, and improved battery pack design to enhance drone safety and reliability.

Project Repository

GitHub Repository: https://github.com/madhukarkalyan40-dev/drone-battery-thermal-instability-analysis

References

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[3] A. Pesaran, "Battery thermal models for hybrid vehicles."

[4] M. Doyle, "Design and simulation of lithium-ion battery systems."

[5] P. Ramadass, "Modeling of lithium-ion batteries for electric vehicles."